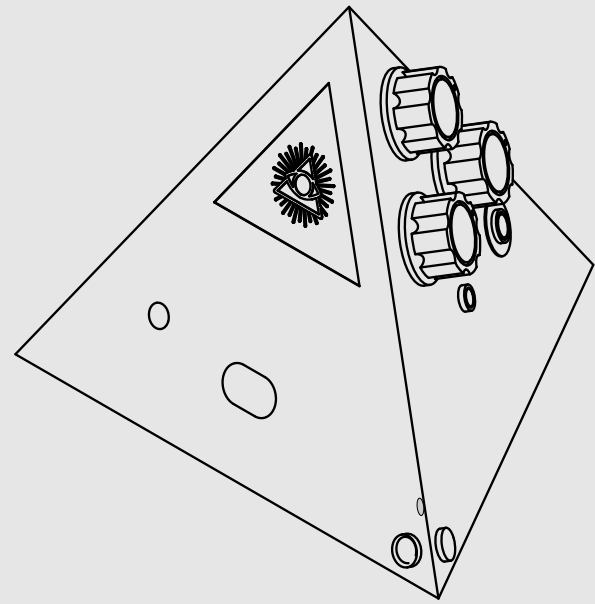


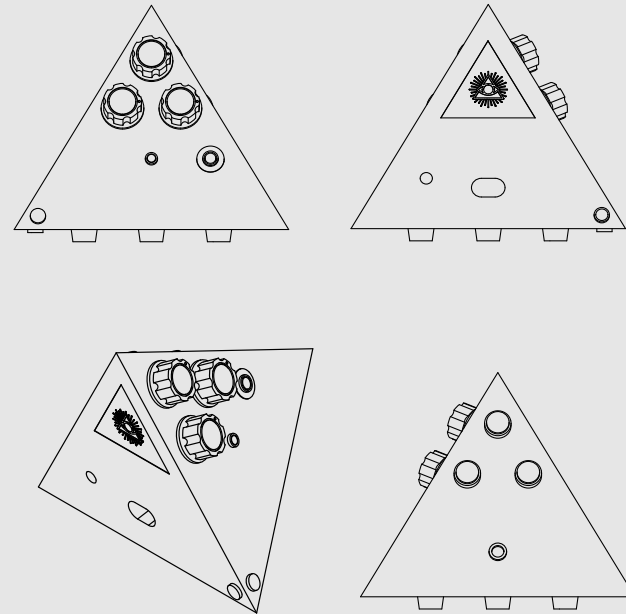
Agnostic box



Manual

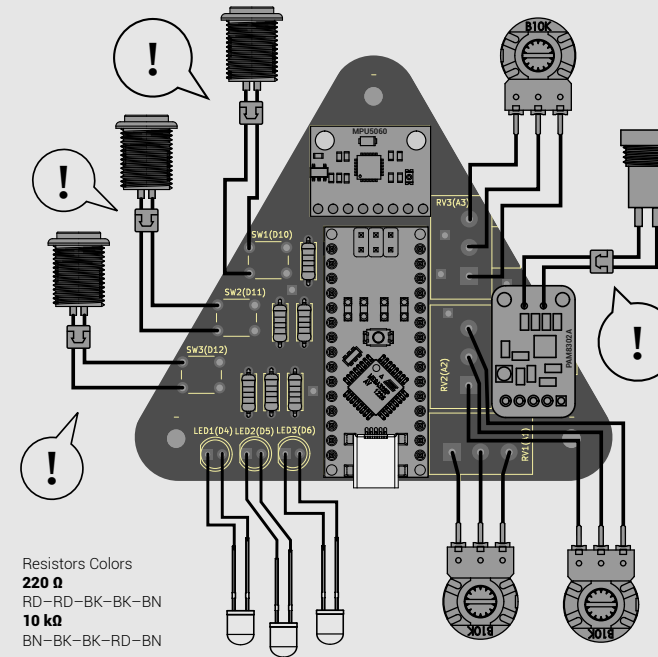
Asnapshot!

The Agnostic Box is a completely open source, hackable sonic device designed for research and teaching. All source files and instructions are available in a public GitHub repository.



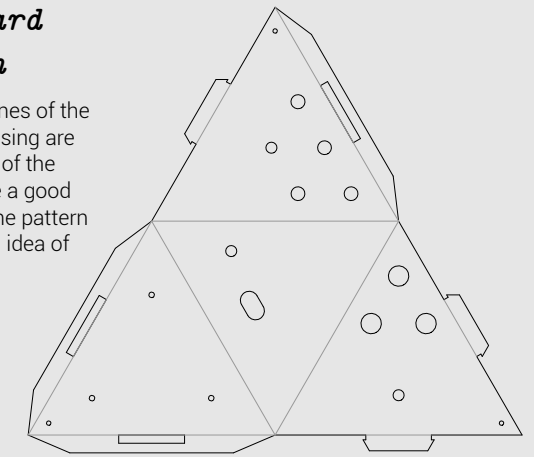
Circuit

Assemble the circuit using a soldering iron. Pay attention to the components that require a connector. The connector is there to allow easier assembly of a number of components. The Arduino, amplifier, and acc/gyro should be placed on the PCB directly, though a connector is strongly recommended so you can eventually remove them and use them in other projects! Note that the electronic components may differ from those shown in the images depending on the manufacturer, but the overall connections should be the same..



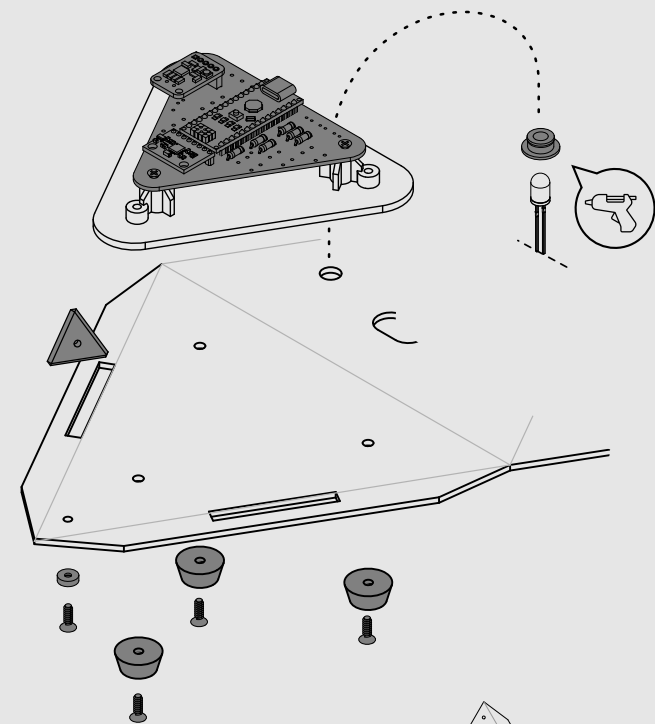
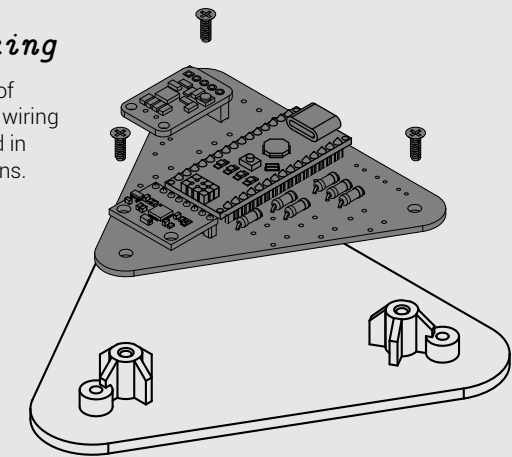
Cardboard pattern

The folding lines of the cardboard casing are on the inside of the box. It can be a good idea to fold the pattern first to get an idea of how it works.

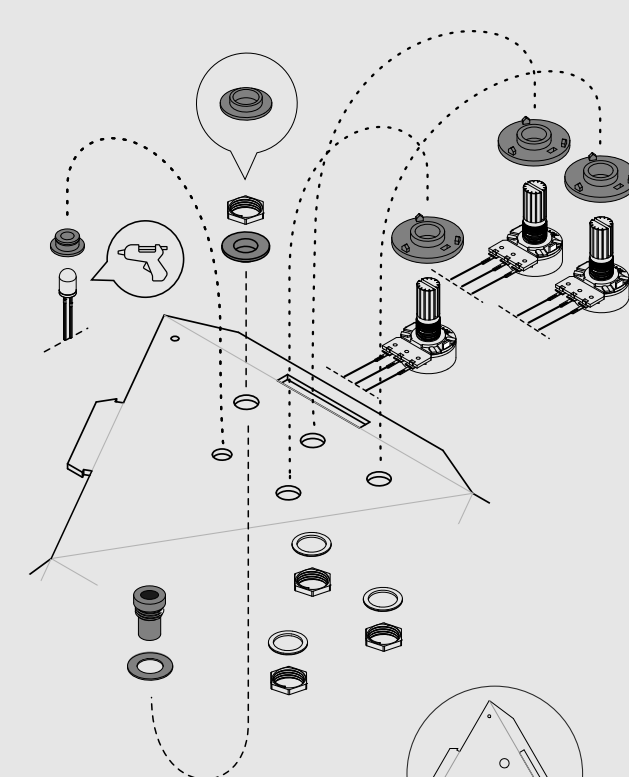
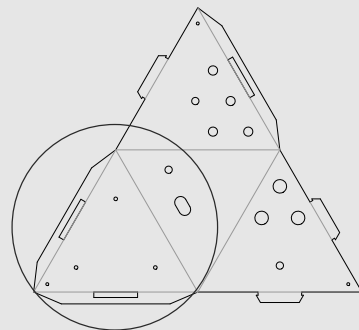


PCB placing

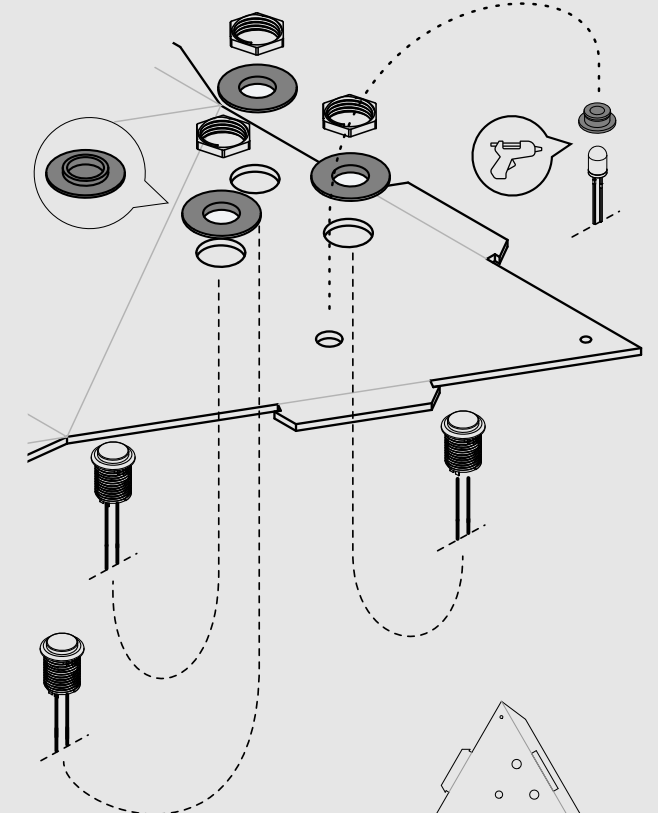
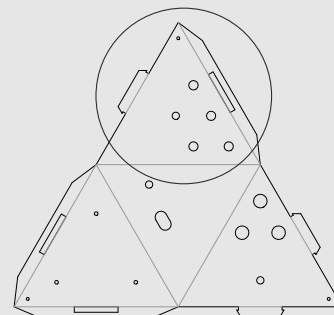
Note: For ease of reading, messy wiring is not visualized in these illustrations.



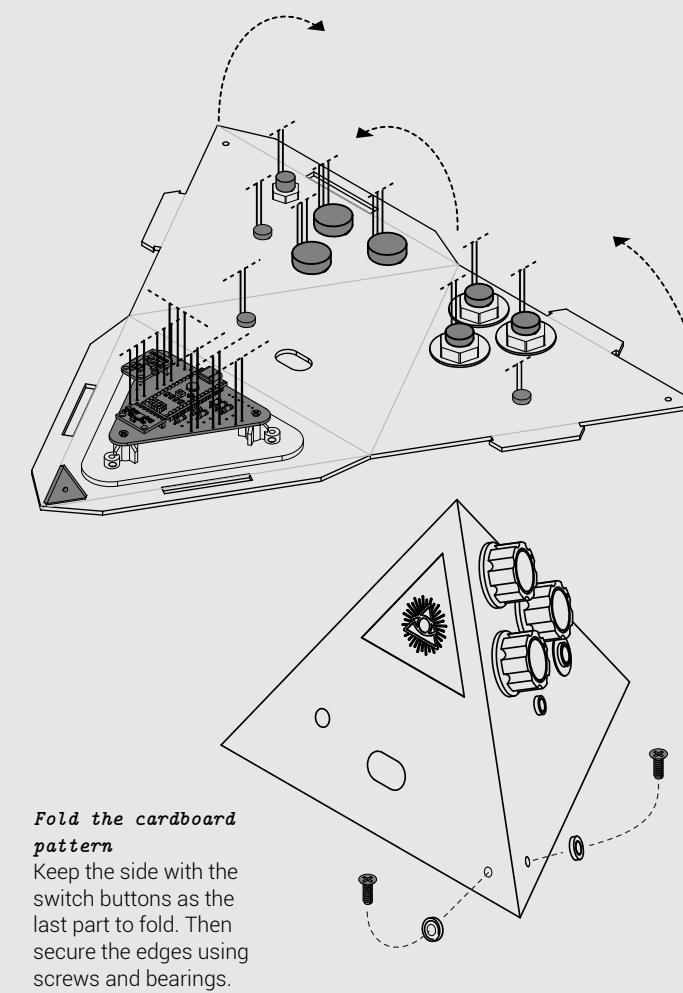
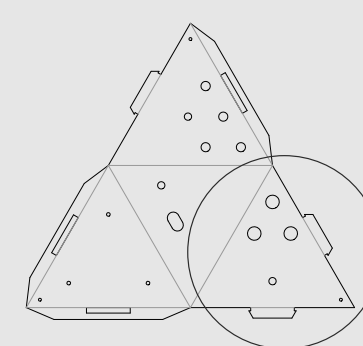
PCB support, casing junction (be gentle on that!), and LED placement
 Use hot glue to secure the LED to the plastic element.



Potentiometers, LED (same as the others), and **audio jack**
 Make sure the orientation of the knobs is correct (Wires perpendicular to the lower side of the face).



Switches and last LED
 Disconnect the switches using the connector, then re-attach them.

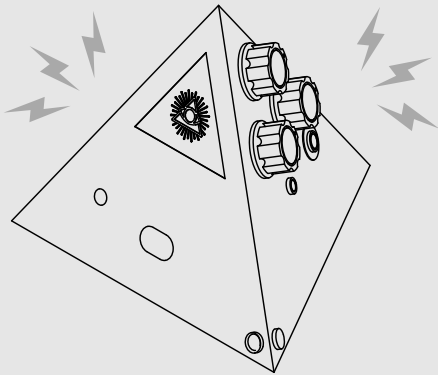


Fold the cardboard pattern
 Keep the side with the switch buttons as the last part to fold. Then secure the edges using screws and bearings.

Meet the Agnostic Box!

The Agnostic Box is a simple and hackable sonic device designed for research and teaching purposes. It contains 3 knobs, 3 buttons, a mono audio out jack, and an MPU6050 that detects acceleration and orientation. The device is driven by an ESP32 Arduino Nano board, but it can also be used with a traditional Arduino Nano. It is extremely versatile and can be programmed in many different ways according to your needs. This device is completely open source. You can find all the source files and instructions in a public repository on GitHub.

By default the agnostic box is programmed with a Dub Siren-ish digital sythn that you can just plug to a speaker, but you can du much more...



Program the Agnostic Box!

What You'll Need

Hardware: Agnostic Box (**Arduino Nano ESP32 inside**) + **USB-C cable**
Software: Arduino IDE 2.x (arduino.cc/en/software)

Step-by-Step

Launch the IDE & Open Board Manager
Open the Arduino IDE. In the left sidebar, click the Boards Manager icon (cube shape).

Install the Board Package

In the search bar, type Nano ESP32. Click Install on the official Arduino ESP32 Boards package.

Connect Your Board

Plug the board into your computer using a USB-C cable.

Select Your Board

Click the board selector dropdown (often "Select Other Board & Port"). Find and select Arduino Nano ESP32. If two versions appear, choose the one labeled USB.

Write & Upload Your Code

Blink test: Go to File -> Examples -> 01.Basics -> Blink.
Compile: Click the checkmark (top-left).

Upload: Click the right arrow next to it.

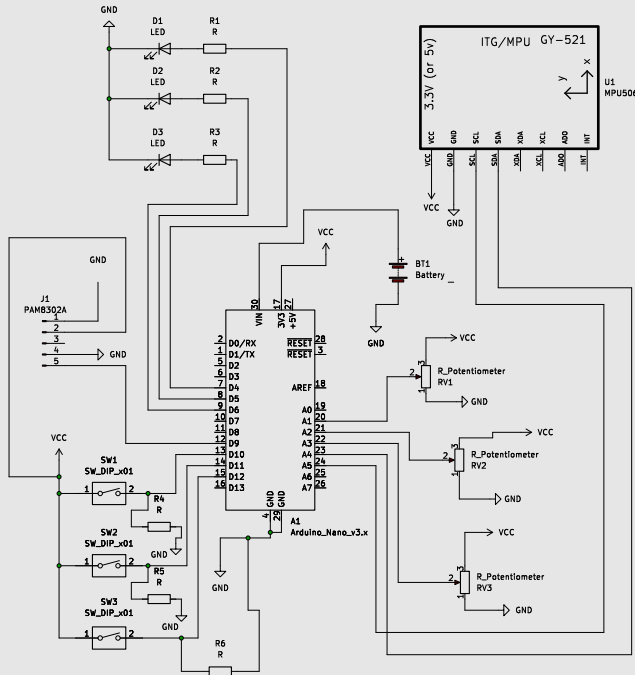
See the result

The built-in RGB LED on the board should start blinking.

Note - if your Agnostic Box is equipped with a classic Arduino Nano (or clone) you don't need any of this (but beware: no bluetooth and no wifi available)

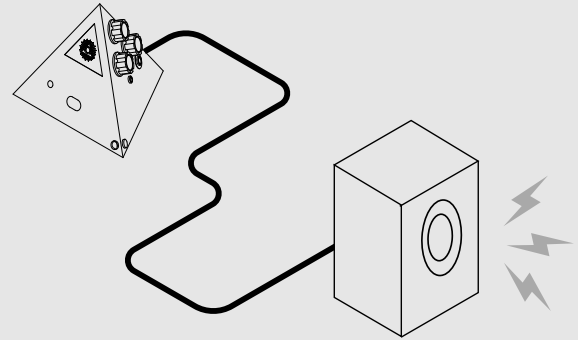
Schematics and pinout

RV1 (10K pot)	RV2 (10K pot)	RV3 (10K pot)	MPU5060
Wiper > A1	Wiper > A2	Wiper > A3	VCC > 3.3V
Outer1 > 3.3V	Outer1 > 3.3V	Outer1 > 3.3V	GND > GND
Outer2 > GND	Outer2 > GND	Outer2 > GND	SCL > A5
			SDA > A4
SW1 > D10	LED1 > D4		
SW2 > D11	LED2 > D5		
SW3 > D12	LED3 > D6...		



Agnostic Box and the Mozzi Library

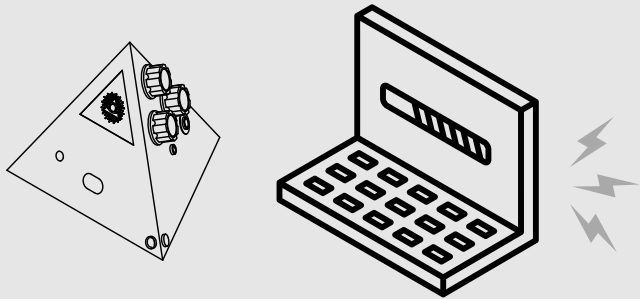
The Mozzi library transforms your Agnostic Box into a powerful, standalone digital synthesizer. It's a sound synthesis library for Arduino that allows you to generate complex audio like growls, sweeps, and chorusing atmospherics using synthesis units like oscillators, delays, filters, and envelopes. For your box, the three knobs could control synthesis parameters such as frequency, filter cutoff, or amplitude. The three buttons could trigger different sound banks or envelopes. The MPU5060 could introduce gesture-controlled pitch bends or vibrato, while the three LEDs can provide real-time visual feedback of the synthesis state. Mozzi is available for the ESP32, making it a perfect fit for the Nano ESP32. The easiest way to get started is to install it directly from the Arduino Library Manager



<https://github.com/sensorium/Mozzi>
<https://sensorium.github.io/Mozzi/>
<https://docs.arduino.cc/hardware/nano-esp32>

Agnostic Box and the BLE MIDI Library

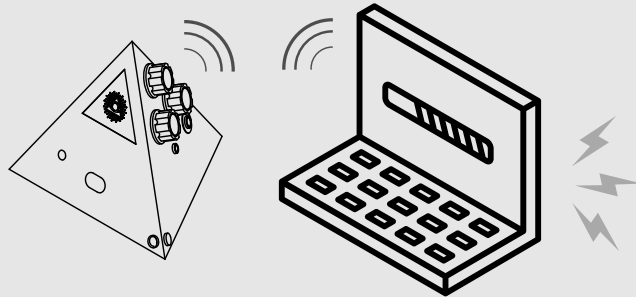
By using the BLE-MIDI protocol, your Agnostic Box becomes a natively recognized MIDI peripheral. This means that once connected, your computer, smartphone, or tablet will see the box as any other commercial Bluetooth MIDI controller—no special drivers or bridge scripts are required. This is because the Arduino Nano ESP32's ESP32-S3 chip, part of the u-blox NORA-W106 module, supports both Wi-Fi and Bluetooth Low Energy (LE). The key detail is that the Nano ESP32 is an ESP32-S3, which, unlike older ESP32 chips, can support MIDI over both BLE and USB simultaneously. In this setup, the three LEDs on your box can even be controlled by incoming MIDI messages from the DAW, creating a fully bidirectional wireless instrument.



<https://www.midi.org/specifications/midi-transport-specifications/midi-over-bluetooth-le>
<https://docs.arduino.cc/hardware/nano-esp32>
<https://github.com/lathoub/arduino-ble-midi>

Agnostic Box and OSC Protocol

Open Sound Control (OSC) offers a more flexible and higher-resolution alternative to MIDI, and with the Nano ESP32's built-in Wi-Fi, you can implement this natively. You can use the MicroOsc library, a simple and lightweight Open Sound Control (OSC) library that supports the ESP32 platform. Your Agnostic Box can act as an OSC client over Wi-Fi, streaming sensor data directly. For instance, the three knobs could send float values as OSC messages like /box/knob1 0.67, and the MPU5060 could send /box/imu/roll 45.2 to a target IP address and port where software like Pure Data or Max/MSP is listening. MicroOsc provides a unified API to work over UDP (Ethernet or WiFi), making it ideal for this application. This approach avoids the need for a computer as a bridge, maintaining a clean and direct wireless connection.



<https://explore.market.dev/ecosystems/arduino/projects/microosc>
<https://github.com/CNMAT/OSC>
<https://docs.arduino.cc/hardware/nano-esp32>

Agnostic Box and whatever

Beyond synthesis and MIDI/OSC control, the Agnostic Box can serve as a multi-purpose musical tool. The combination of knobs, buttons, an IMU, and LEDs makes it ideal for gestural performance: tilt the box to control filter sweeps while pressing buttons to trigger samples. You could use the MPU5060's gyroscope to map bowing or shaking gestures to pitch bends, vibrato depth, or delay feedback. The three LEDs can be programmed as a visual metronome (blinking at BPM-synced rates) or as a feedback display for note velocities. Over Wi-Fi (using the Nano ESP32's built-in radio), the box can stream real-time sensor data to a networked sound server running SuperCollider or Pure Data. Alternatively, use the board's USB port to emulate a class-compliant MIDI device—plugging directly into a computer without any driver installation. With the onboard memory, you could even store short wavetables or drum samples, turning the box into a minimal groovebox when combined with Mozzi's sample playback features. The same three buttons could be repurposed as a step sequencer interface: each button toggles a step on/off, while knobs adjust pitch or velocity per step, and the LEDs light up to show the current playhead position. In an educational context, the box becomes a tangible platform for teaching synthesis, mapping, and sensor-to-sound relationships—all without soldering or external hardware beyond the Arduino Nano ESP32 itself.

Credits

The agnostic box concept comes from an original idea from Raul Masu and Nicolo Merendino (aka "Chi ha Ucciso Il Conte?")

Design of the box and of this manual is by Nicolo Merendino (aka "Chi ha Ucciso Il Conte?")

Project funded by the HKUST-GZ

More info - chihauccisoilconte.eu